ANAMOLY DETECTION IN MEDICAL IMAGES USING AI AND ENSURING DATA PRIVACY

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Abstact - This paper demonstrates the practical implementation of an AI-based anomaly detection system for medical images such as MRI, CT, and X-ray scans. The implemented system is designed to support healthcare professionals by automatically detecting abnormal patterns that may indicate potential diseases, reducing dependency on manual image analysis. CNNs are implemented for extracting meaningful spatial features from medical images, and autoencoders learn normal image representations and detect anomalies based on reconstruction error. The critical issue of maintaining patient data confidentiality is addressed through the incorporation of privacy-preserving techniques, including Differential Privacy, Federated Learning, and Secure Multi-Party Computation into the training and inference process. Explainable AI techniques such as heatmaps and saliency maps enhance the system's transparency and clinical trust. Finally, the solution will be deployed as a secure web-based application, enabling real-time analysis, visualization, and alert generation while ensuring data privacy and security

Keywords— Medical Image Anomaly Detection, Convolutional Neural Networks, Autoencoders, Privacy-Preserving AI, Differential Privacy, Federated Learning, Secure Multi-Party Computation, Explainable AI, Healthcare Security

I. INTRODUCTION

The healthcare sector heavily relies on imaging methods, such as Magnetic Resonance Imaging (MRI), Computed Tomography (CT) and X-ray imaging, to visualize the inside of a person. These forms of imaging are prevalent throughout the healthcare system and assist in both diagnosing illness and monitoring recovery. Currently, however, the traditional model for assessing such images includes a radiologist using manual interpretation to evaluate each image. This analysis can be slow and prone to human error. Additionally, there are increasing numbers of images generated in the healthcare system over time, leading to increased workloads for healthcare professionals. Additionally, recently developed artificial intelligence (AI) and deep learning technologies have made great strides in automating the analysis of medical images. Deep learning methodologies, including convolutional neural networks (CNNs) and autoencoders, are well-suited for extracting complex features and identifying abnormal patterns from medical images. These systems traditionally use training datasets to learn about normal anatomical structures in a way that will enable them to detect abnormalities even when there is limited labeled data regarding these abnormalities.

Though these above-mentioned benefits can be achieved, there are many serious concerns related to implementing AI technology in the healthcare sector—especially around patient data privacy and confidentiality. Medical images contain sensitive information, and thus need to follow rigorous privacy regulations. As a result of centralised model training for AI systems, there exists an inherent risk of breaches and the need for innovative new methods that will help protect patients' privacy within the context of the implementation of an anomaly detection system.

II. SYSTEM WORKFLOW

The fact that the aforementioned benefits can be realized through the application of AI systems in healthcare is tempered by the fact that there are serious issues regarding the implementation of AI systems into the healthcare/medical field regarding patient's right to privacy and confidentiality of their data. Medical images typically contain sensitive information about an individual, and therefore are subject to strict regulations related to their use. Since the models for machine learning are centralized, there is potential for data breaches. Accordingly, new innovative methodologies must be employed; therefore, because there are new methodologies used in this model rection method for patient privacy and confidentiality purposes.

A carefully designed model has been implemented to automate the identification of anomalous medical imaging, while safeguarding the information associated with those images. Our workflow begins with the acquisition of medical images (CT, MRI, etc.) from both publicly accessible repositories and medical facilities. Pre-processing (such as image rescaling, normalizing, denoising, and data augmenting) is performed on the acquired medical images to standardize their quality, permitting the models to learn effectively and ensuring that identifying characteristics of the individuals depicted in the images are eliminated.

With a view to minimizing privacy concerns while providing clinicians with easier access to use, the solution has added privacy-preserving and interpretability modules into the system. In the aforementioned solution, privacy is safeguarded during training and cooperative use through methods of Differential Privacy, Federated Learning and Secure MultiParty Computation so that none of the patient's personal information will be released. Additionally, heat map and saliency map techniques are used as part of the explainable component of the application to indicate to a clinician which area contains the abnormally proportioned part of an image. Ultimately all final results will be produced via a secured webinterface system.

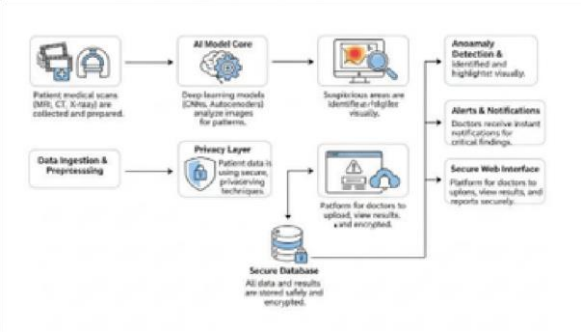


Figure 1. System Workflow

II. DATASET COLLECTION AND PREPROCESSING

This Study will utilize medical imaging using the MRI, CT and X-Ray from publicly available and benchmarked medical imaging repositories (i.e. both normal and abnormal case examples) in a research environment to conduct an analysis of medical images. Medical images are grouped according to the type of imaging technique so that the current study may analyze the anomaly detection method independently and in combination. The data for this Study will not contain any identifying or personally identifiable medical patient data.

Figure 2. Dataset collection and preprocessing

To improve image quality and create consistency among all datasets, preprocessing is conducted through an extensive pipeline. The images will be resized to the same resolution (fixed) before starting to build the model, which means that the input size of the data being supplied will be consistent throughout development. Images will have all intensity (brightness and contrast) differences between the scans normalized, to produce standard visual results. Noise will be removed through various noise-reduction techniques so that unnecessary items do not create "artifacts". Image rotations/flips/scalings will be used as augmentation techniques to increase the robustness of the model and reduce the possibility of overfitting. In addition to conducting the visual preprocessing procedures previously described, the various types of medical images are handled with care to prevent any possible privacy issues.

Figure 3. Healthy CT Scan of a Brain and chest

Figure 4.Un-Healthy CT Scan of a Brain and chest

IV. CNN-BASED FEATURE EXTRACTION

The new design of the system uses the Convolutional Neural Networks (CNN) for feature extraction from medical images. CNN's are especially helpful in analysing tasks associated with medical images since they can learn the spatial hierarchy; local traits like edges, texture, and shape are automatically learned by the system! After pre-processing the medical images, they are input to the CNN model with learnable filters in the convolution layers. The CNN architecture consists of multiple convolutional layers, activation functions and pooling layers. While the first few convolutional layers will begin to extract low-level features like edge locations and intensity change, the deeper convolutional layers will extract the requisite high-level anatomical features needed to diagnose ailments using medical imaging. Pooling layers will also reduce the dimensionality of the data while preserving useful information and maintain a hierarchical structure so the CNN model can correctly identified all normal and abnormal medications based on their anatomy. The feature maps generated by CNN are effective representations of the input image that will continue to be used in the autoencoder based anomalies detection module for future training processes and creation of new and improved detection methods. Decoupling the functions of feature extraction and anomaly detection will reduce time spent training data used for creating an anomaly detection algorithm because there are fewer interactions that must be trained.

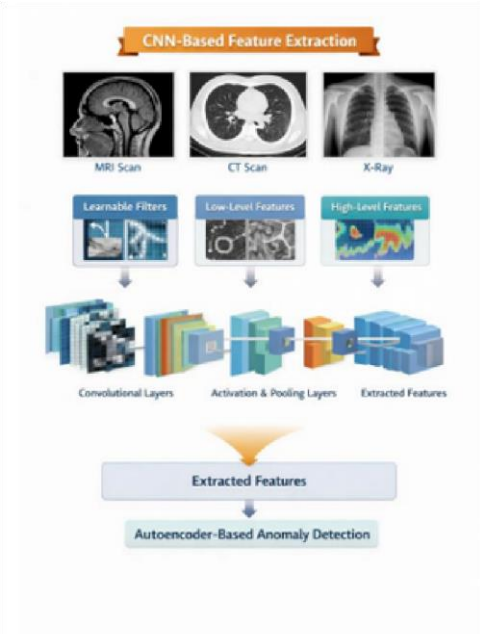


Figure 5. Cnn-Based Feature Extraction

V. AUTOENCODER-BASED ANOMALY DETECTION

An autoencoder is used as the anomaly detection module in this system. The autoencoder is comprised of an encoder and a decoder that comprise a neural network and learn from input data (i.e., an image) by learning the normal patterns of a given type of medical image. Autoencoders fall in the category of unsupervised neural networks. The encoder compresses the incoming image (the encoded representation of an image) down to a low-dimensional latent representation. The decoder attempts to create a close approximation of the incoming image from the low-dimensional latent representation. The majority of the autoencoder's initial training is conducted with normal MRI's, CT's, and X-ray images to enable the model to understand the actual anatomical structure of typical healthy medical images. The autoencoder learns to minimize its reconstruction loss (the difference between the reconstructed and original images)

Figure 6. Autoencoder-Based Anomaly Detection

The proposed method relies on an autoencoder for robust anomaly detection where there are not enough anomalous examples to enable reliable detection. A combination of CNN features paired with the autoencoder increases reliable detection. Unsupervised detection is supported by this architecture making it very suitable for real-life healthcare situations where there are not enough abnormal samples available to train models properly. The significance of the autoencoder in identifying the existence o

VI. PRIVACY-PRESERVING FRAMEWORK IMPLEMENTATION

The preservation of patient privacy is very important in medical image analysis. This framework not only provides a means to analyse anomalies in a way that protects the privacy of medical images, but also makes sure that sensitive patient information contained within medical images remains confidential. Since medical images contain a wide range of information about patients, unauthorised access to that information as well as leaking that information has serious ethical and legal consequences. In order to protect patientidentifying information, the system also has multiple mechanisms that can be combined to protect patient data and still maintain a high level of model performance. One of these ways is to apply Differential Privacy in the training phase of the model, that is adding noise to either the gradients or the model updates, that helps to ensure that the model's parameters cannot be traced back to any specific patient record, even when someone has access to the model's parameters. The ability to trace a parameter back to a specific record is based on the amount of privacy budget used to create a balance between the privacy of the data versus the accuracy of the detection.

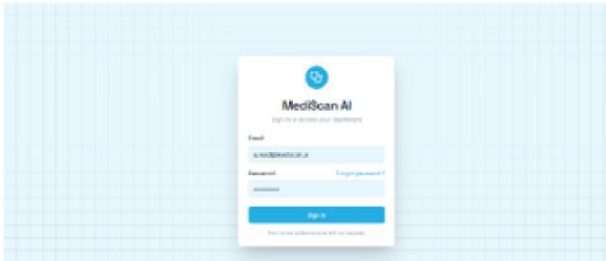


Figure 7. Login Page

Apart from Differential Privacy, Federated Learning and SMPC (i.e., Secure Multi-Party Computation) are other technologies that have been integrated into the platform to allow multiple institutions to share compute resources for collaborative model training. Federated Learning allows institutions to perform model training on data stored locally by each institution, so that data does not leave its original source, but aggregating encrypted model updates for use in the aggregate model. The use of Secure Multi-Party Computation provides additional assurance of the system's security by requiring that at all times, all intermediate steps involved in the aggregation and collaborative analysis of models are performed on encrypted data. SMPC provides a cryptographic protocol to ensure that no data or sensitive information can be accessed by any participating institution, including the aggregation server, during the aggregation process. Therefore, by providing at least three different data privacy methods (i.e., Differential Privacy, Federated Learning, and SMPC), the platform creates a comprehensive, robust, and multi-layered data privacy framework.

VII. EXPLAINABLE AI AND VISUALIZATION

In the healthcare industry, explainability is a fundamental aspect of AI systems because physicians rely on these AI-based tools for their interpretation of results generated by these tools; therefore, they must be able to understand how these systems generate their results. This project implements explainable AI techniques by providing visual information regarding the anomalies detected in medical imaging. Instead of solely providing numeric outputs for each anomaly or a number signifying its likelihood of being an anomaly, each anomaly detected by the AI is visually represented to the physician by showing the location within the scanned images (i.e., MRI, CT, and X-ray) of the specific areas of the image analyzed by the AI. Gradient-weighted Class Activation Mapping (Grad-CAM) is a technique used to create heat maps indicating the location of the pixels within the analyzed scan that had a substantial effect on the classification of an image as an anomaly; this technique is a function of the gradients that are derived from the convolutional neural network (CNN) layers that were evaluated during the classification of the image and allows the evaluators to focus on the pixels

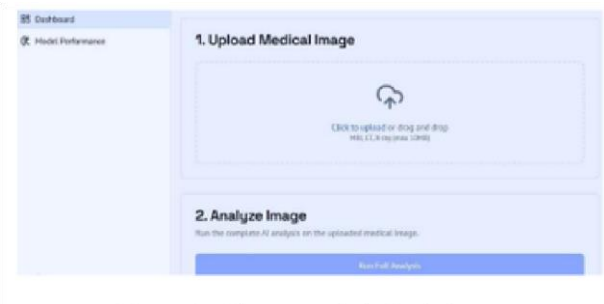


Figure 8. Threat Analysis Techniques

The system user is then presented with both heat maps and anomaly maps in the system interface, along with their corresponding medical images, allowing radiologists and other healthcare providers to validate their own findings against the AI explanation of an anomaly detection result. By including an explanation of AI methods used in an anomaly detection system, the system provides an automated method of assessing medical images as well as an adjunct tool for physicians to use when making decisions regarding patient care.

VIII. ALERT GENERATION AND WEBBASED DEPLOYMENT

The automated alert system provides healthcare personnel with timely notifications about suspected abnormalities in their medical image scans. An abnormal scan will be identified through the reconstruction error and assessment score thresholds determined by the autoencoder. Once identified as a probable abnormality through the above methods, the alert generation process starts, intending to send a notification of the image's potential danger to the medical personnel as soon as possible. The alert system relies on defined assessment score thresholds through experimentation; therefore, if the score of the detected abnormal image is higher than the defined threshold, alerts will be generated containing information, such as scan type, confidence level, and scan date/time. Severityrelated categories (less severe to more severe) may be assigned to alerts to prioritise the degree of concern for potential danger in any medical images.

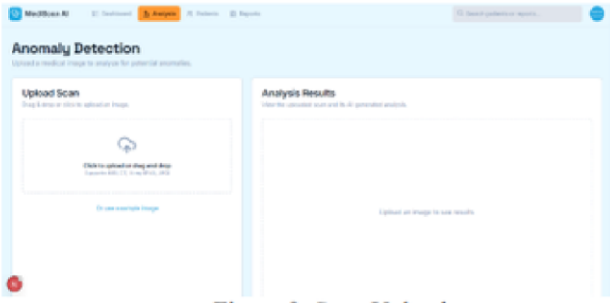


Figure 9. Scan Upload

The anomaly detection system has been developed into a secure web application utilizing the Django or Flask framework to improve usability of the anomaly detection system. The web application allows doctors, technicians and administrators to upload medical images, view results of the anomaly detection, and create explainable AI visualizations. Access control has been implemented to ensure that medical images are only available to authorized users, which increases the overall security of the system. The anomaly detection system is easily integrated

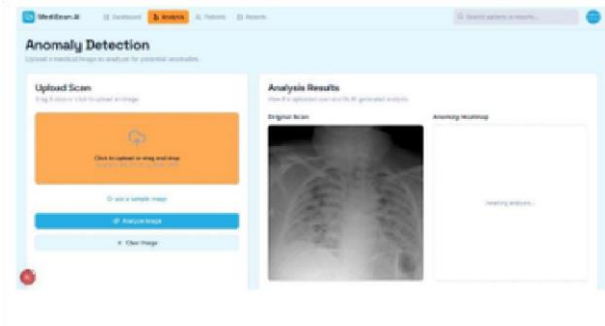


Figure 10. Uploaded Scan

The web application has built-in features that allow safe storage and retrieval of data as well as the generation of reports. You can ensure that detailed anonymized data can be kept securely for use later on, heat maps and anomaly score will be available for reference as well.Additionally, the architecture of the system is available for expansion (via scalable solutions) and can also integrate with existing hospital information systems or cloud platforms if desired. A finished solution will link artificial intelligence, anomaly detection, automated alerts

IX. PERFORMANCE EVALUATION METRICS

Commonly accepted metrics within the field of medical image analysis will be used to evaluate performance for the deployed anomaly detection system. Anomaly detection in medical imaging has a very high sensitivity and specificity; therefore, multiple metrics will be used to evaluate performance versus just one metric alone. These metrics provide a comprehensive evaluation of the system's ability to detect anomalies and to do so with a minimal amount of false alarms. The system evaluation will take into consideration the importance of a medical diagnosis, where both false negatives and false positives can have serious implications. Accuracy is the first metric that will be used to evaluate the system's overall effectiveness in classifying medical images as being either normal or abnormal. However, accuracy alone is not adequate in medical applications.

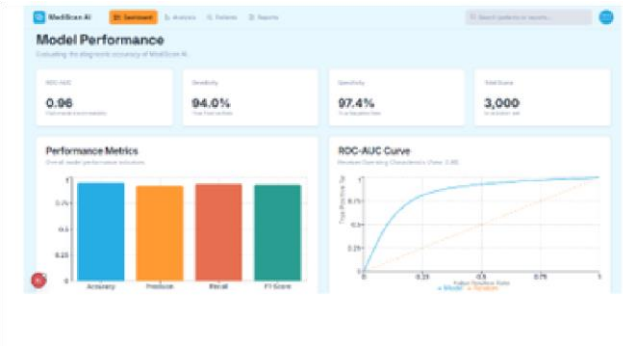


Figure 11. Model and Accuracy

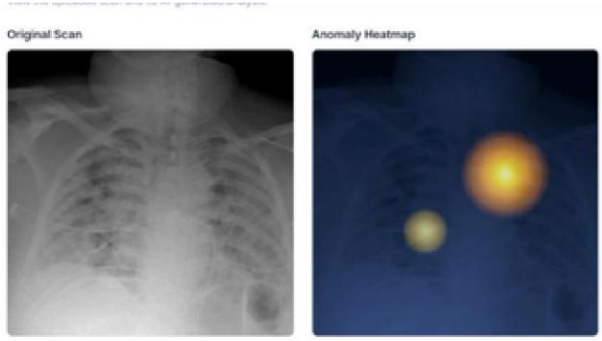
The F1 score is used as a measure of the accuracy vs. completeness tradeoff, providing a single number to reflect the overall detection ability of the system. Two additional measures, sensitivity and specificity, provide an indication of the diagnostic accuracy of the system. Sensitivity is used to measure how effectively the system can detect that an image contains abnormalities. On the other hand, specificity is used to measure whether the system is able to determine that an image does NOT contain any abnormalities. The Receiver Operating Characteristic curve (ROC) and Area Under the Curve (AUC) are also used to evaluate system performance.

The relationship between the actual true positive and False Positive will be plotted over a range of different threshold levels, The higher the AUC (Area Under The Curve) of the ROC curve, the better the accuracy of the system in distinguishing normal and abnormal images. An acceptable

Figure 12. Confusion Matrix

ROC AUC was produced across the MRI, CT and X-ray datasets.

Overall, our analysis demonstrates that our evaluation criteria demonstrate that our solution provides an accurate and efficient method of detecting anomalies, while still being clinically relevant. Additionally, the combination of CNNbased feature extraction with autoencoder-based reconstruction and privacy-preserving training makes for a system with stable and reliable outputs. As a result, we believe our solution can effectively support healthcare providers in delivering accurate and privacy-protective diagnostic support.

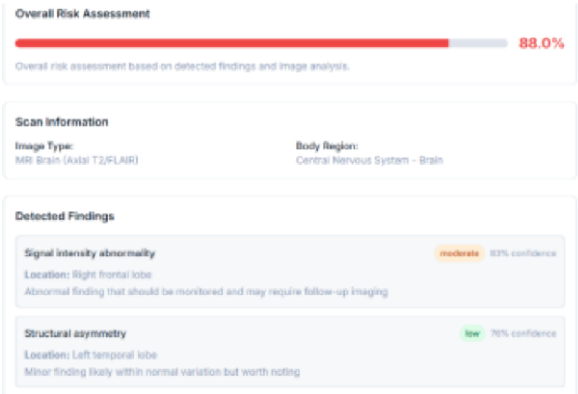


X. COMPARATIVE ANALYSIS AND SYSTEM EFFECTIVENESS

There is an evaluation of the efficacy of the new system with traditional systems for medical image analysis. The traditional systems mostly use manual or simple operations such as thresholding and edge detection. These systems do not work well when trying to identify complex patterns in the body and small abnormalities; therefore, they give unreliable diagnostic test results. However, the AI-based system creates its own features automatically.

Figure 13. Heatmap anomaly Detection

Incorporating CNN-based features into an anomaly detection system increases detection rates significantly compared with pixel-only methods of identifying anomalies such as raw data, pixel values, labels or indicators; etc. A convolutional neural



network provides the capability to extract useful spatial and structural features from the inputs to the anomaly detection module that improve the quality of those inputs leading to reducing erroneously identified anomalies or accurate localization of identified anomalies. In addition, the combined use of Convolutional Neural Networks and Autoencoders yields superior performances compared to either Convolutional Neural Networks or Autoencoders used separately. Another key differentiator between the proposed approach and similar systems is the degree of protection afforded to users' private information.

Figure 14. Overall Risk Assessment

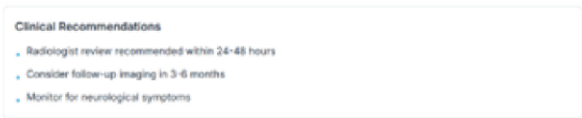


Figure 15. Clinical Recommendation

The system's use of visible AI-sourced visualizations and web deployment allow for a high degree of usability, with regard to

the eventual adoption and trust in AI systems by physicians. The AI provides heat maps to visualize the areas of the medical images being examined and saliency maps to display the areas of interest for the physician, allowing them to understand and gain confidence in the AI's decision-making. The automated notifications and certificate-based access further improve the usability of the system. The comparative analysis between the proposed system and traditional and existing medical image analysis systems

XI. LIMITATIONS

While the implemented anomaly detection system demonstrates strong performance, there are also global limitations discovered throughout development. The key found limitation is that the model's success depends heavily upon the training dataset quality. Due to variations in the acquisition resolution and imaging modality for medical images, these datasets can be substantially different from one another. In turn, this variability may lead to challenges with generalization of the resultant model or to the need for additional fine-tuning if the model is applied to an alternative site. Additionally, while the unsupervised learning model (i.e. autoencoders) used for the anomaly detection system is very effective for use with limited or non-existent labeled data associated with areas of abnormality, there are some instances where normal anatomical variations may erroneously

Heatmaps and saliency maps, commonly referred to as explainability methods for AI clinicians, are designed to help improve the transparency of the models. In some instances these techniques may not be able to provide accurate clinical explanations. Other times, the areas identified by the model

The current model relies heavily on image-level anomaly detection to find anomalies but does not perform classification of disease or provide any segmentation of the area where the disease has occurred. The lack of accurate localization and labeling of a disease may serve as a disadvantage when it comes to the potential future clinical uses of this system, as it will be hard to provide an accurate patient diagnosis without an accurate localization

XII. RESULTS AND DISCUSSION

An AI based anomaly detection system has been tested for efficacy and reliability using MRI, CT, and X-ray medical images. Its results showed that it is capable of accurately detecting both normal and abnormal images through reconstruction error-based scoring methodology. Additionally, the use of convolutional neural networks (CNNs) as a feature extraction method significantly improved the representation learned by autoencoders, enabling even the smallest structural anomalies to be detected which would be difficult to detect through visual inspection alone. The data produced in these experiments demonstrate the ability of the system to generalize across all modalities of medical imaging. Standard metrics used in this study to evaluate performance included accuracy, precision, recall, F1 Score, sensitivity, specificity, and ROCAUC. Sensitivity was measured to be high indicating that the system has the ability to find abnormal medical images effectively; therefore, These are critical requirements of a clinical diagnostic tool in order to provide accurate diagnoses. Other metrics related to specificity remained stable indicating that the system also has the ability to identify normal medical images accurately without generating fals

Based on reconstruction error, the autoencoder demonstrated a clear separation of class (normal vs abnormal) with high errors for abnormal and low for the class of normal images. The use of CNN for the extraction of image features reduced noise and instability in the training of autoencoders using raw images, thus increasing the accuracy of the model's ability to detect anomalies. Methods for preserving privacy were tested for their effect on the model's ability to operate correctly.

The importance of explainable AI visualization in validating model predictions is paramount. Heatmaps and saliency maps effectively emphasized areas of clinical importance corresponding to the identified anomalies. The visual explanation provided healthcare professionals insight into the AI predictions. Therefore, these results indicate that the developed model has a balance among accuracy, privacy, and interpretability.

XIII. CONCLUSION

This project illustrates a machine learning based approach to valuable insights from healthcare image data. Our robot has learned the distinction between normal and abnormal patterns in MRI, CT and X-ray images using the combination of convolutional neural networks to extract features from image data and autoencoders to detect anomalies in a supervised manner. In doing so, we were able to employ an automated solution that required minimal human intervention to evaluate medical imaging and provide assistance to clinicians to achieve faster and more accurate diagnoses. The implementation of differential privacy, federated learning, and secure multi-party computation demonstrates that it is possible to secure patient data throughout the sampling, training and evaluation of the AI anomalies while also having a valid method to protect patient data that would otherwise impact the performance of the AI systems. Furthermore, our creation of heat maps and saliency maps can be used as a visual explanation to provide meaningful transparency when making decisions based upon AI algorithms and automated notifications provided a mechanism for timely identification of possible high-risk patients. Overall, our work provides a scalable, secure, and interpretable anomaly detection approach to medical imaging that has the potential for real world application within the healthcare domain as AI continues to evolve.

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